Analyzing Short-term, Solar Activity Bursts: Exploring the Capability for Predicting Next Burst's Properties

Ainsley Helgerson[1,2], Mausumi Dikpati[2], Kiran Jain[3], Breno Raphaldini[2]

[1] Embry-Riddle Aeronautical University, Daytona Beach, FL, United States, [2] NSF NCAR, High Altitude Observatory, Boulder, CO, United States,
[3] National Solar Observatory, Boulder, CO, United States

~ Introduction ~

The Sun experiences an 11-year solar cycle. Within these cycles, there are short-term quasi-annual variabilities, which are fluctuations in the common pattern seen in a Sun cycle. There are two types of short-term variabilities, one is called the Reiger type, which lasts around 6 – 18 months. The second type of burst is called a Quasi-Biennial variation, lasting around 1.5 – 2 years.

We are statistically analyzing long-term solar activity data to derive the properties of upcoming bursts.

Keywords: Sunspot Number (SSN), Short-term variability, Carrington Rotations (CR)

~ Methods ~

We executed this analysis using the following steps:

1. Analysis was performed using **SILSO daily data**, which includes the total, northern, and southern hemispheres.
2. Running averages with **Gaussian smoothing** were used to derive the short-term variability.
3. Subtracted the **mean** or basal solar cycle trend from **smoothed daily SSN**.
4. Derived the **peak amplitude** and **full width at half maximum** (FWHM) of all these bursts, then determined the corresponding **periods** of these bursts.
5. Used **Lag correlation** between the periods and amplitudes of each burst. **Autocorrelation** was also used for each property. Both hemispheres were calculated.
6. Interpreted the **results** of lag and **autocorrelation** to find a relationship of predicting next burst's properties.

~ SILSO Daily Data and Smoothed Carrington Rotations (Steps 1 & 2) ~

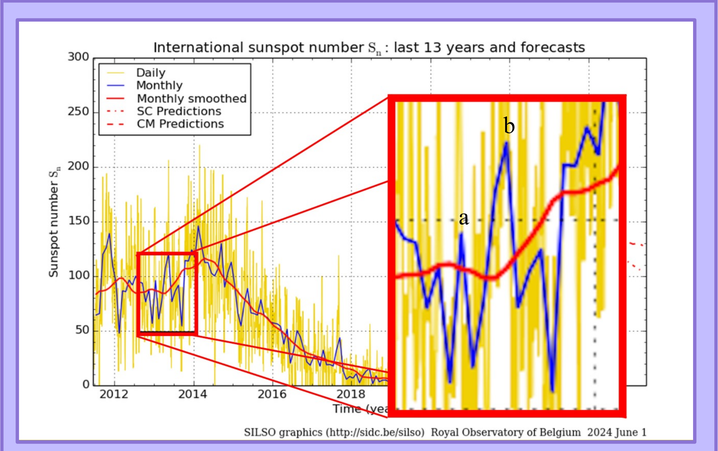

Fig.1. Sunspot Number (S_n) v. Years (2011 – 2024)
- **Blue**: Monthly nonhemispheric sunspot number
- **Red**: Monthly nonhemispheric smoothed sunspot number

Step 1: The zoomed in figure displays an example of a short period burst (a), which is then followed by a high amplitude burst (b). This pattern shows up consistently within the monthly hemispheric data (blue). The results from our analysis can be used to build a forecasting tool which will assess the nature of upcoming enhanced bursts of activity.

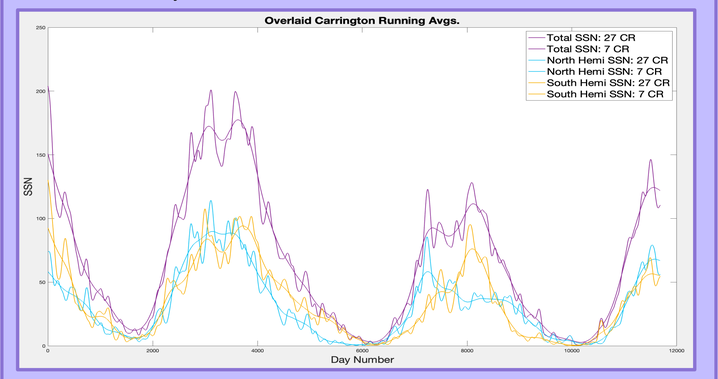

Fig.2. SSN v. Day Counts (1/1/1992 – Present)
- **Purple**: Total hemispheric daily data presented as 7 CR and 27 CR running averages with Gaussian smoothing
- **Blue**: Northern hemispheric daily data (same conditions)
- **Yellow**: Southern hemispheric daily data (same conditions)

Step 2: Running averages with Gaussian smoothing of 5, 7, 9, and 27 were calculated using daily hemispheric sunspot number data. By doing this, the appropriate average was made to achieve the greatest accuracy and precision in our results.

Acknowledgements: This work is supported by the NSF National Center for Atmospheric Research, which is a major facility sponsored by the National Science Foundation under cooperative agreement 1852977. We acknowledge support from COFFIES Phase II NASA-DRIVE Center for the subaward 80NSSC22M0162 from Stanford. This research was supported by the National Science Foundation REU program, Award #1950911.

~ Detrended Solar Cycles, Periods (FWHM) and Amplitudes (Steps 3 & 4) ~

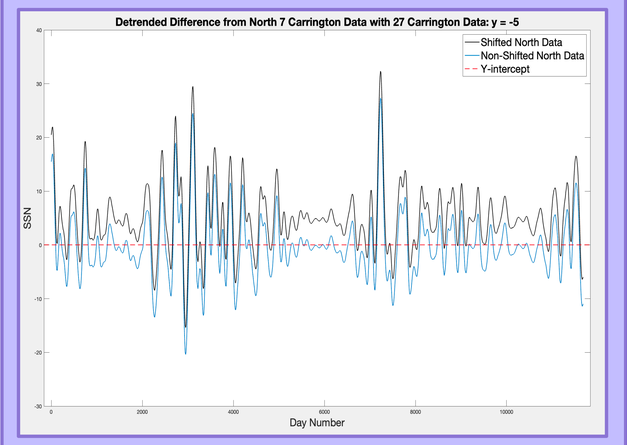

Fig.3. SSN v. Detrended Day Counts (1/1/1992 – Present)
- **Black**: Shifted northern hemispheric data
- **Blue**: Non-shifted northern hemispheric data
- **Red**: Y-Intercept

Step 3: The detrended dataset was derived by subtracting the 27 CR curve from the 7 CR curve. This was done to analyze each burst without the shifts in the 11-year solar cycle. The largest fluctuations in the current results are related to the shift in the y-intercept. This constraint causes all values below the y-intercept line to be neglected. Therefore, the data must be shifted to account for larger bursts.

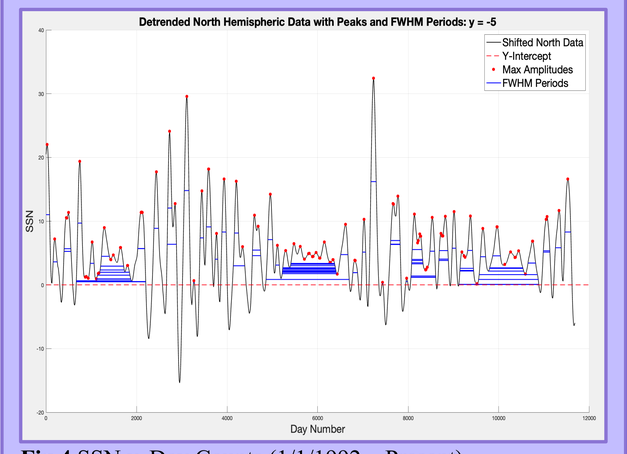

Fig.4 SSN v. Day Counts (1/1/1992 – Present)
- **Black**: Shifted northern hemispheric data
- **Blue**: FWHM values
- **Red**: Corresponding maximum amplitudes

Step 4: The number of periods and amplitudes changed based on the placement of the y-intercept. Once these properties were found, we then were able to perform multiple analyses to determine correlations.

~ Interpreting Lag Correlation and Autocorrelation (Steps 5 & 6) ~

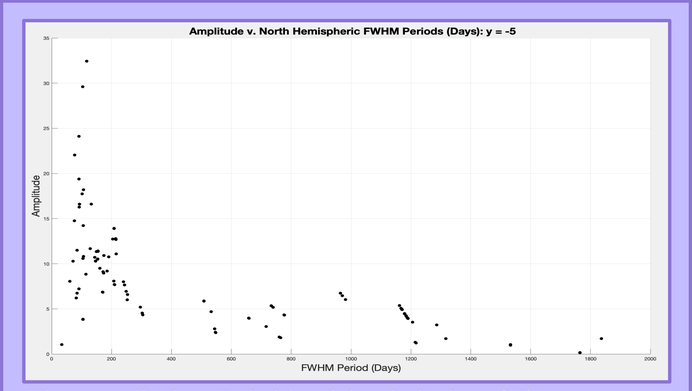

Fig.5 Amplitude v. Period (north hemispheric: 7 CR)

Steps 5 & 6: This scatter plot represents a visual relationship of lag correlation. Our results for this case (northern hemispheric data: 7 CR) includes:
- Moderate inverse (lag) correlation **(~ 50%)** between the period of the *(n)-th* burst and the amplitude of the *(n+1)-th* burst, implying that the shorter the burst duration, the stronger the next burst is.
- Very strong autocorrelation **(~ 72%)** among the periods of the bursts. This indicates that the Sun has a short-term memory of 1-2 years, and hence the periods from one burst to the next are not random but follow a pattern.

~ Summary & Concluding Remarks ~

Biggest takeaways:
- **~ 50%** Inverse (lag) correlation
- **~ 72%** Autocorrelation

Next steps:
- Use extended hemispheric sunspot number data
- Exclude solar minimum: code is already developed
- Use long-term data from the Solar Cycle Science website (http://solarcyclescience.com/activeregions.html)

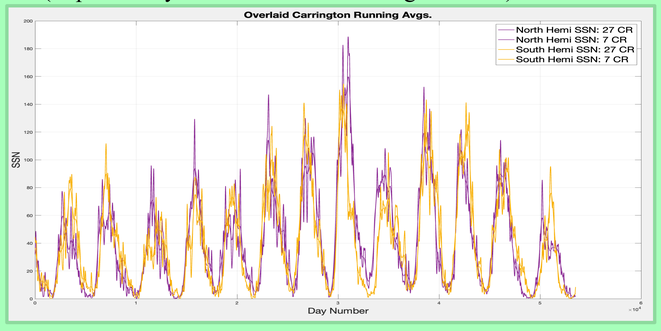

Fig.6 SSN v. Day Counts (05/01/1874 – 10/31/2020)
- **Purple**: Extended north hemispheric data (7 CR)
- **Yellow**: Extended south hemispheric data (7 CR)

This specific case (extended northern hemispheric data: 7 CR), resulted in ~ 82% autocorrelation and ~ 32% lag correlation. All other cases will be derived in the upcoming months.